Haotian Liu

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EDUCATION

Xiamen University (XMU), Xiamen, China Sep. 2023 – Jun. 2027 (expected)

B.Eng. in Software Engineering | Average: 82.66/100

Relevant Coursework:
Machine Learning for Decision-Making and Optimization (96); Algorithms for Modern AI (94); Introduction to Artificial Intelligence (93); Interactive Design (90); The Practice of Computer Programming (92).

PUBLICATIONS

PilotBench: A Benchmark for General Aviation Agents.
IJCNN 2026, **accepted**. Co-first author. arXiv:2604.08987

FLY-EVAL++: An Evidence-Driven Evaluation Protocol for Flight Prediction with Large Language Models.
COLM 2026, **accepted**. arXiv:2609.04021

Beyond Answer Voting: Fine-Grained Correction of LLM Reasoning through Graph Alignment across Sampled Traces.
ARR, under review. First author. arXiv:2610.18606

LFUSE: Learned Evidence Fusion for Noisy Spoken and Image-Grounded Product Search.
ICASSP 2027, under review. First author.

Before the Agent Acts: Understanding, Task Accounts, and Implementation in Pre-Coding Review.
EMSE, under review. First author.

RESEARCH PROJECTS

DDRG: Disagreement-Driven Verification Guided by Reasoning Graphs Apr. 2026 – Present

First Author | Research with Dr. Zihui (Irene) Li | Matsuo–Iwasawa Lab, The University of Tokyo

- Proposed DDRG, which aligns source-grounded claim graphs across sampled reasoning traces and triggers one focused repair only when an answer-blind verifier refutes the majority answer's supporting path.
- Improved macro-average accuracy over direct answering by 8.65 and 15.83 points for Qwen2.5-7B and Llama3.1-8B across five benchmarks, exceeding CoT, GoT, GoV, and self-consistency.

CETI: Before the Agent Acts Apr. 2026 – Present

First Author | ZGC Lab

- Proposed CETI, a repository-grounded pre-coding review workflow that represents task constraints as editable **Records** linked to pinned source passages, allowing developers to revise an agent's task interpretation before code generation.
- Evaluated with 24 participants across 72 repository-task cases (+17.2-point adjusted understanding contrast vs. FreeChat-2) and traced request families through 504 coding attempts on 10 tasks and 3 models.

Long-Horizon Memory Governance for LLM Agents (LHMG) Dec. 2025 – Apr. 2026

First Author | School of Data Science, Fudan University

- Developed a conflict-aware memory-governance framework for long-horizon LLM agents, combining revision graphs, No-Silent-Overwrite updates, risk-gated conditioning, and governed forgetting.
- Observed a 13.3-point GPT-4o gain in cross-turn decision consistency over flat memory in controlled diagnostics.

PilotBench: General Aviation Agent Benchmark Dec. 2025 – Mar. 2026

Core Contributor | School of Data Science, Fudan University

- Co-developed PilotBench (708 real trajectories, 34-channel telemetry, nine flight phases); evaluated 41 models.
- Found a precision–controllability split: forecasters reached MAE 7.01; LLMs, 86–89% instruction following at 11–14 MAE.

NeuroMark: Multimodal Reliability Evaluation for Neuropsychiatry Sep. 2025 – Mar. 2026

Research Contributor | School of Data Science, Fudan University

- Contributed to a benchmark of clinical reasoning and brain-imaging interpretation (2,379 questions, six imaging tasks); evaluated 20 LLMs and 16 VLMs and helped develop the SCR, EA, DDI, and JTI metrics.

INDUSTRY EXPERIENCE

LFUSE: Learned Evidence Fusion for Procurement Search Nov. 2025 – Apr. 2026

First Author | H World Group

- Developed LFUSE for ProcuraClaw, a learned evidence-fusion ranker that reranks BM25 and dense candidates using rank, character, pinyin, and query-agreement features, with noise-aware training for corrupted and ASR queries.
- Improved MRR@10 over tuned convex fusion by 7.8% on average across three human-judged Chinese Multi-CPR domains while reducing fusion-stage latency by 80–100× versus LLM-based adaptive fusion; transferred to a 1,854-SKU hotel catalog.

RESEARCH AND TECHNICAL SKILLS

AI and Programming: LLM agents and memory, multimodal learning, AI for software engineering and HCI, RAG and information retrieval, benchmark and evaluation design; Python, PyTorch, Hugging Face, Spring Boot, SQL, Java/Java EE, C/C++, Unity.
Research Methods and Tools: Controlled experiments, human studies, ablation and error analysis; Git, Linux, Docker, LaTeX.
Languages and Tests: Chinese (native); English (TOEFL iBT 90/120); Japanese (reading knowledge).